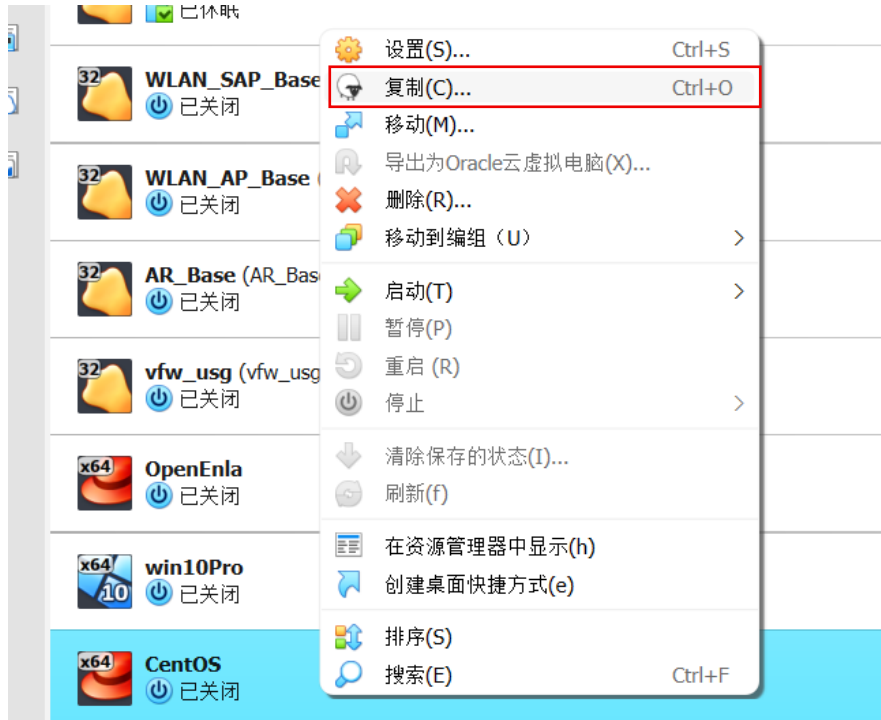


集群化软件安装配置前准备

一、进行主机复制（在 VMware 叫克隆）



其他设置保持一致

已经完成三个节点的复制



二、进行基本网络配置

1. 下一步进行节点 IP 和主机名的修改，将 node1 的 IP 地址改为 192.168.56.110，主机名为 node1，将 node2 的 IP 地址改为 192.168.56.120，主机名为 node2。将 node3 的 IP 地址改为 192.168.56.130，主机名为 node3

以 node2 为例，在 root 权限下，静态修改主机的网络配置和主机名

```
密码：
[root@node2 xcx]# nmcli connection modify "hostonly-56" \
> ipv4.addresses 192.168.56.120/24 \
> ipv4.gateway 192.168.56.1 \
> ipv4.dns "223.5.5.5" \
> ipv4.method manual
[root@node2 xcx]# nmcli connection up "hostonly-56"
[root@node2 xcx]# hostnamectl set-hostname node2
[root@node2 xcx]# hostname
node2
[root@node2 xcx]#
```

连接远程终端 FinalShell，查看 ip

```
[xcx@node2 ~]$
[xcx@node2 ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe8b:774b prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8b:77:4b txqueuelen 1000 (Ethernet)
    RX packets 140 bytes 15013 (14.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 216 bytes 22621 (22.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.120 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a738:6602:e201:2aaa prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:99:01:15 txqueuelen 1000 (Ethernet)
    RX packets 44967 bytes 3732405 (3.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 77281 bytes 15609576 (14.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

已经修改完成

2.验证网络连通性

```
已发送 2 个包, 已接收 2 个包, 0% packet loss, time 1017ms
rtt min/avg/max/mdev = 0.297/0.370/0.444/0.073 ms
[root@node1 ~]#
[root@node1 ~]# ping 192.168.56.110
PING 192.168.56.110 (192.168.56.110) 56(84) 比特的数据。
64 比特, 来自 192.168.56.110: icmp_seq=1 ttl=64 时间=0.121 毫秒
^C
--- 192.168.56.110 ping 统计 ---
已发送 1 个包, 已接收 1 个包, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.121/0.121/0.121/0.000 ms
[root@node1 ~]# ping 192.168.56.120
PING 192.168.56.120 (192.168.56.120) 56(84) 比特的数据。
64 比特, 来自 192.168.56.120: icmp_seq=1 ttl=64 时间=0.286 毫秒
^C
--- 192.168.56.120 ping 统计 ---
已发送 1 个包, 已接收 1 个包, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.286/0.286/0.286/0.000 ms
[root@node1 ~]#
[root@node1 ~]# ping 192.168.56.130
PING 192.168.56.130 (192.168.56.130) 56(84) 比特的数据。
64 比特, 来自 192.168.56.130: icmp_seq=1 ttl=64 时间=0.274 毫秒
```

Node1 可以 ping 通包括自己的所有节点

3.建立 IP 和主机名映射

进入每个服务器的/etc/hosts 文件, **都添加下面三条信息**

```
[root@node1 ~]# vim /etc/hosts
[root@node1 ~]#

127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

192.168.56.110 node1
192.168.56.120 node2
192.168.56.130 node3

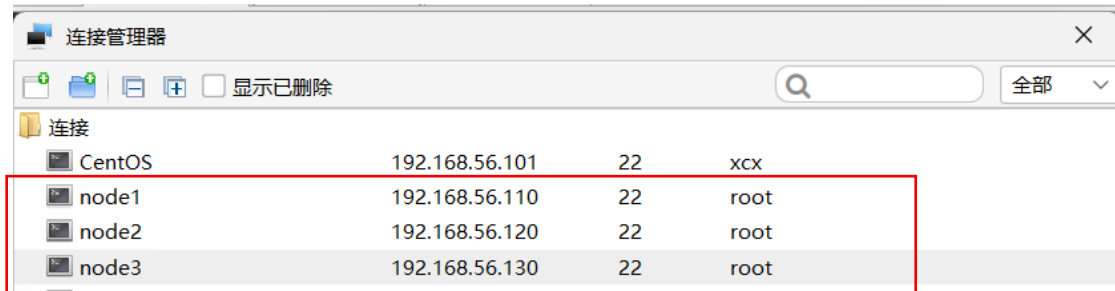
~
~
~
~
```

直接 ping 主机名,

```
[root@node1 ~]# ping node1
PING node1 (192.168.56.110) 56(84) 比特的数据。
64 比特, 来自 node1 (192.168.56.110): icmp_seq=1 ttl=64 时间=0.020 毫秒
64 比特, 来自 node1 (192.168.56.110): icmp_seq=2 ttl=64 时间=0.028 毫秒
^C
--- node1 ping 统计 ---
已发送 2 个包, 已接收 2 个包, 0% packet loss, time 1030ms
rtt min/avg/max/mdev = 0.020/0.024/0.028/0.004 ms
[root@node1 ~]# ping node2
PING node2 (192.168.56.120) 56(84) 比特的数据。
64 比特, 来自 node2 (192.168.56.120): icmp_seq=1 ttl=64 时间=0.284 毫秒
^C
--- node2 ping 统计 ---
已发送 1 个包, 已接收 1 个包, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.284/0.284/0.284/0.000 ms
[root@node1 ~]# ping node3
PING node3 (192.168.56.130) 56(84) 比特的数据。
64 比特, 来自 node3 (192.168.56.130): icmp_seq=1 ttl=64 时间=0.363 毫秒
```

也可以完成连通性测试

在 Finalshell 中已经建立起 node1 和 node2 的远程连接



连接	IP	端口	用户名
CentOS	192.168.56.101	22	xcx
node1	192.168.56.110	22	root
node2	192.168.56.120	22	root
node3	192.168.56.130	22	root

三、配置 SSH 免密登录

以 node1 登录 node3 为例

```
[root@node1 ~]# ssh root@node3
The authenticity of host 'node3 (192.168.56.130)' can't be established.
ED25519 key fingerprint is SHA256:v+ApTcw8RkvVQV4pr+1VgYVu1CeSWSM6oa0jI1PCodA.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: node2
  ~/.ssh/known_hosts:4: node1
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'node3' (ED25519) to the list of known hosts.
root@node3's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sat Apr 25 16:24:07 2026 from 192.168.56.1
[root@node3 ~]#
```

已经进入 node3 服务器

由于进行集群实验, 要频繁远程登录其他服务器, 避免每次登录都要输入密码, 我们采取 SSH 免密登录, 核心就是配置一个公钥, 让每个服务器都拥有进入其他服务器的钥匙

操作如下

输入这条指令, 为服务器创建一个远程登录的钥匙

```
[root@node3 ~]# ssh-keygen -t rsa -b 4096
```

把密钥分享给 node1,node2,node3(全部节点), 每个服务器都要把密钥分享给包括自己的所有节点

执行下面三条指令

```
ssh-copy-id node1
ssh-copy-id node2
ssh-copy-id node3
```

```
[root@node3 ~]# ssh-copy-id node1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host 'node1 (192.168.56.110)' can't be established.
ED25519 key fingerprint is SHA256:v+ApTcw8RkvVQV4pr+1VgYVu1CeSWSM6oa0jI1PCodA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'node1' (ED25519) to the list of known hosts.
root@node1's password:

[root@node3 ~]# ssh-copy-id node2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host 'node2 (192.168.56.120)' can't be established.
ED25519 key fingerprint is SHA256:v+ApTcw8RkvVQV4pr+1VgYVu1CeSWSM6oa0jI1PCodA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'node2' (ED25519) to the list of known hosts.
root@node2's password:

[root@node3 ~]# ssh-copy-id node3
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host 'node3 (192.168.56.130)' can't be established.
ED25519 key fingerprint is SHA256:v+ApTcw8RkvVQV4pr+1VgYVu1CeSWSM6oa0jI1PCodA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'node3' (ED25519) to the list of known hosts.
root@node3's password:
```

直接使用 SSH 登录，不需要输入密码即可登录

```
[root@node3 ~]# ssh node1
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sat Apr 25 16:24:02 2026 from 192.168.56.1
[root@node1 ~]#
```

四、关闭防火墙和 SELinux

因为三个节点之间的通信通过端口访问，关闭防火墙有利于后续实验
每一个节点都进行如下操作

```
[root@node1 ~]# systemctl stop firewalld
[root@node1 ~]# systemctl disable firewalld
[root@node1 ~]# systemctl status firewalld
o firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)
[root@node1 ~]#
```

进入/etc/sysconfig/selinux 文件，将 enforcing 修改为 disabled

```
#
# grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#   targeted - Targeted processes are protected,
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

```
21 #
22 SELINUX=disabled
23 # SELINUXTYPE= can take one of these three values:
24 #   targeted - Targeted processes are protected,
25 #   mls - Multi Level Security protection.
26 SELINUXTYPE=targeted
27
28
:
```

至此集群化部署的实验准备已全部完成，可以重启虚拟机开始新的集群实验